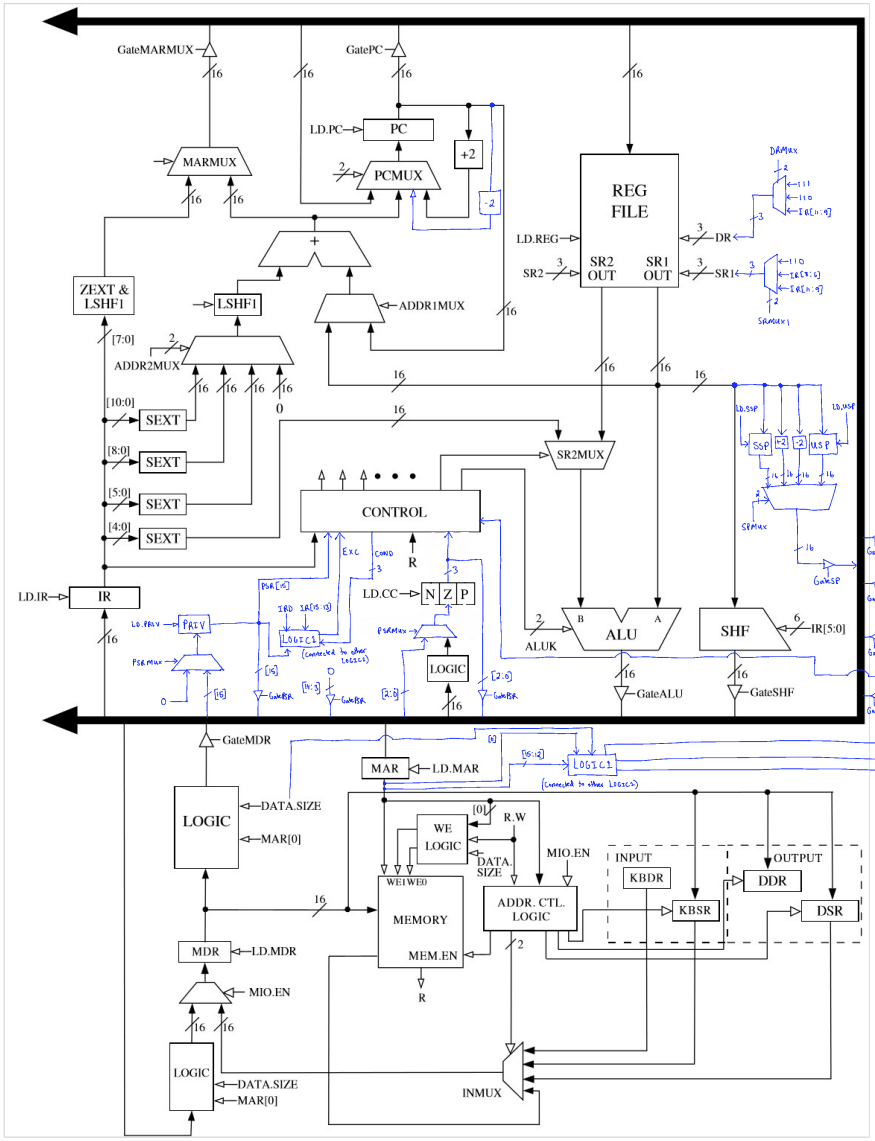
datapath

Sunday, November 1, 2026 8:07 PM

datapath



Muxes	00	01	10	11
PCMux	PC+2	Bus	ADDR	PC-2
DRMux	IR[11:9]	R7	R6	
SR1Mux	IR[11:9]	IR[8:6]	R6	
SPMux	R6+2	R6-2	USP	SSP
ExcMux	INTV	x02	x03	x04

PSR Mux

PSRMux = 0 : PRIV ← 0 , NEP ← logic
PSRMux = 1 : PRIV ← Bus[15] , NEP ← Bus[2:0]

LOGIC1

unal : Checks (DATA_SIZE) & (MAR[0]) — Checking for unaligned access

priv : (PRIV) & (~MAR[15]) & (~MAR[14]) & (~(MAR[15] & MAR[14])) — checking if in user mode and MAR is below x3000

exc : priv | unal - checking for exception for micro sequencer ld1

unk : (IRD) & (IR[15]) & (~IR[14]) & (IR[13]) - checking for unknown opcode

int : INTV ≠ 0 - checking if there's an interrupt vector waiting

LD.Excv : ((cond == 101) & int) | ((cond == 110) & exc)) | unk — loading only when something needs to be so that it can hold value across states

CLR.INTV : (cond == 101) & int - clears interrupt upon acknowledgement

ExcMux[1] : unk | ((cond == 110) & (~priv))

ExcMux[0] : unk | ((cond == 110) & (priv))